In the following exercises, all coordinates and components are given with respect to an orthonormal frame $\mathcal{K} = (O, \mathcal{B})$. For the 2-dimensional cases $\mathcal{B} = (\mathbf{i}, \mathbf{j})$ and for the 3-dimensional cases $\mathcal{B} = (\mathbf{i}, \mathbf{j}, \mathbf{k})$.

1. (1 point) Select all that apply. Which of the following are direct isometries of $\mathbb{E}^3$?

   A. glide-reflections B. glide-rotations C. rotation-reflections D. translations

2. (1 point) Let $\phi(\mathbf{x}) = A\mathbf{x} + b$ be an indirect isometry of $\mathbb{E}^3$. Which of the following are true? (Select all that apply)

   A. A admits -1 as an eigenvalue
   B. $A \in \mathrm{O}(3)$
   C. b is an eigenvector for A
   D. $A \in \mathrm{SO}(3)$

Setup 1: In $\mathbb{E}^3$, consider the point $P(3,2,0)$, the vector $v(0,1,1)$, the plane $\pi : y - 1 = 0$, the orthogonal reflection $\mathrm{Ref}_{\pi}^{\perp}$ in π, and the parallel reflection $\mathrm{Ref}_{\pi,v}$ in π along v. Moreover, let ϕ be the transformation obtained by applying $\mathrm{Ref}_{\pi}^{\perp}$ after $\mathrm{Ref}_{\pi,v}$, i.e.,

$$\phi = \mathrm{Ref}_{\pi}^{\perp} \circ \mathrm{Ref}_{\pi,v}.$$

3. (1 point) Consider Setup 1 above. The coordinates of the point $\phi(P)$ are __________.

4. (1 point) Consider Setup 1 above. The map ϕ is

   A. a direct isometry B. an indirect isometry C. not an isometry D. a rotation

5. (1 point) Consider Setup 1 above. Then $\phi\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} \_ & \_ & \_ \\ \_ & \_ & \_ \\ \_ & \_ & \_ \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \begin{bmatrix} \_ \\ \_ \\ \_ \end{bmatrix}.$

Setup 2: In $\mathbb{E}^3$, consider the point $P(1,2,1)$, the vector $v(0,1,1)$, the plane $\pi : z - 1 = 0$, the parallel reflection $\mathrm{Ref}_{\pi,v}$ in π along v, and the standard rotation $\mathrm{Rot}_{Ox,45^\circ}$ around the x-axis by an angle of 45°. Moreover, let ϕ be the transformation obtained by applying $\mathrm{Ref}_{\pi,v}$ after $\mathrm{Rot}_{Ox,45^\circ}$, i.e.,

$$\phi = \mathrm{Ref}_{\pi,v} \circ \mathrm{Rot}_{Ox,45^\circ}.$$

6. (1 point) Consider Setup 2 above. The coordinates of the point $\phi(P)$ are __________.

7. (1 point) Consider Setup 2 above. The map ϕ is

   A. a direct isometry B. an indirect isometry C. not an isometry D. a rotation

8. (1 point) Consider Setup 2 above. Then $\phi\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} \_ & \_ & \_ \\ \_ & \_ & \_ \\ \_ & \_ & \_ \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \begin{bmatrix} \_ \\ \_ \\ \_ \end{bmatrix}.$

9. (1 point) In $\mathbb{E}^3$, consider the orthogonal reflection

$$R\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \frac{1}{9}\begin{bmatrix} 7 & -4 & 4 \\ -4 & 1 & 8 \\ 4 & 8 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}.$$

   An equation of the reflection plane of the reflection R is ____________.

Answers:

1. B, D
2. A, B
3. $(3, 2, -2)$
4. C
5. $\phi(\begin{bmatrix} x \\ y \\ z \end{bmatrix}) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$
6. $\left(1, 2 - \frac{5\sqrt{2}}{2}, 2 - \frac{3\sqrt{2}}{2}\right)$
7. C
8. $\phi(\begin{bmatrix} x \\ y \\ z \end{bmatrix}) = \frac{\sqrt{2}}{2} \begin{bmatrix} \sqrt{2} & 0 & 0 \\ 0 & -1 & -3 \\ 0 & -1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix}$
9. $x + 2y - 2z = 0$